KAIZEN MATHS

Lesson starter

Remainder and Factor Theorem

Synthetic Division with Linear Divisors

Name _____

S1

Divide $P(x) = x^2 + x$ by $x + 1$.

S2

Divide $P(x) = x^2 + 2x - 11$ by $x + 5$.

S3

Divide $P(x) = 3x^3 + 14x^2 + 10x + 8$ by $x + 4$.

S4

Divide $P(x) = 3x^3 - x^2 - 4$ by $x - 1$.

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Exit ticket

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E1

Divide $P(x) = 2x^2 + 9x + 4$ by $x + 4$.

E2

Divide $P(x) = 2x^2 - 3x - 10$ by $x + 1$.

E3

Divide $P(x) = 4x^3 - 4x^2 - 4x - 8$ by $x - 2$.

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S1

Divide $P(x) = x^2 + x$ by $x + 1$.

Answer:

$x^2 + x = (x + 1)(x)$

Working:

Step 1: Convert the divisor to its root. The divisor is x + 1, so set it equal to zero and get $x = -1$.

Step 2: Write the coefficients of $P(x) = x^2 + x$ in descending powers of x. Include any zero coefficients if a power is missing.

Step 3: Carry out synthetic division: bring the first coefficient down, multiply by the root, then add down the next column.

-1	1	1	0
		-1	0
	1	0	0

Step 4: The final entry in the bottom row is the remainder. The other bottom-row entries are the quotient coefficients.

Step 5: Write the quotient with one lower degree than the original polynomial.

$Q(x) = x, \quad R = 0$

Because the remainder is 0, the divisor is a factor of the polynomial.

S2

Divide $P(x) = x^2 + 2x - 11$ by $x + 5$.

Answer:

$x^2 + 2x - 11 = (x + 5)(x - 3) + 4$

Working:

Step 1: Convert the divisor to its root. The divisor is x + 5, so set it equal to zero and get $x = -5$.

Step 2: Write the coefficients of $P(x) = x^2 + 2x - 11$ in descending powers of x. Include any zero coefficients if a power is missing.

Step 3: Carry out synthetic division: bring the first coefficient down, multiply by the root, then add down the next column.

-5	1	2	-11
		-5	15
	1	-3	4

Step 4: The final entry in the bottom row is the remainder. The other bottom-row entries are the quotient coefficients.

Step 5: Write the quotient with one lower degree than the original polynomial.

$Q(x) = x - 3, \quad R = 4$

Because the remainder is not 0, the divisor is not a factor of the polynomial.

Teacher answers & working

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S3

Divide $P(x) = 3x^3 + 14x^2 + 10x + 8$ by $x + 4$.

Answer:

$3x^3 + 14x^2 + 10x + 8 = (x + 4)\left(3x^2 + 2x + 2\right)$

Working:

Step 1: Convert the divisor to its root. The divisor is x + 4, so set it equal to zero and get $x = -4$.

Step 2: Write the coefficients of $P(x) = 3x^3 + 14x^2 + 10x + 8$ in descending powers of x. Include any zero coefficients if a power is missing.

Step 3: Carry out synthetic division: bring the first coefficient down, multiply by the root, then add down the next column.

-4	3	14	10	8
		-12	-8	-8
	3	2	2	0

Step 4: The final entry in the bottom row is the remainder. The other bottom-row entries are the quotient coefficients.

Step 5: Write the quotient with one lower degree than the original polynomial.

$Q(x) = 3x^2 + 2x + 2, \quad R = 0$

Because the remainder is 0, the divisor is a factor of the polynomial.

S4

Divide $P(x) = 3x^3 - x^2 - 4$ by $x - 1$.

Answer:

$3x^3 - x^2 - 4 = (x - 1)\left(3x^2 + 2x + 2\right) + -2$

Working:

Step 1: Convert the divisor to its root. The divisor is x - 1, so set it equal to zero and get $x = 1$.

Step 2: Write the coefficients of $P(x) = 3x^3 - x^2 - 4$ in descending powers of x. Include any zero coefficients if a power is missing.

Step 3: Carry out synthetic division: bring the first coefficient down, multiply by the root, then add down the next column.

1	3	-1	0	-4
		3	2	2
	3	2	2	-2

Step 4: The final entry in the bottom row is the remainder. The other bottom-row entries are the quotient coefficients.

Step 5: Write the quotient with one lower degree than the original polynomial.

$Q(x) = 3x^2 + 2x + 2, \quad R = -2$

Because the remainder is not 0, the divisor is not a factor of the polynomial.

Teacher answers & working

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E1

Divide $P(x) = 2x^2 + 9x + 4$ by $x + 4$.

Answer:

$2x^2 + 9x + 4 = (x + 4)(2x + 1)$

Working:

Step 1: Convert the divisor to its root. The divisor is x + 4, so set it equal to zero and get $x = -4$.

Step 2: Write the coefficients of $P(x) = 2x^2 + 9x + 4$ in descending powers of x. Include any zero coefficients if a power is missing.

Step 3: Carry out synthetic division: bring the first coefficient down, multiply by the root, then add down the next column.

-4	2	9	4
		-8	-4
	2	1	0

Step 4: The final entry in the bottom row is the remainder. The other bottom-row entries are the quotient coefficients.

Step 5: Write the quotient with one lower degree than the original polynomial.

$Q(x) = 2x + 1, \quad R = 0$

Because the remainder is 0, the divisor is a factor of the polynomial.

E2

Divide $P(x) = 2x^2 - 3x - 10$ by $x + 1$.

Answer:

$2x^2 - 3x - 10 = (x + 1)(2x - 5) + -5$

Working:

Step 1: Convert the divisor to its root. The divisor is x + 1, so set it equal to zero and get $x = -1$.

Step 2: Write the coefficients of $P(x) = 2x^2 - 3x - 10$ in descending powers of x. Include any zero coefficients if a power is missing.

Step 3: Carry out synthetic division: bring the first coefficient down, multiply by the root, then add down the next column.

-1	2	-3	-10
		-2	5
	2	-5	-5

Step 4: The final entry in the bottom row is the remainder. The other bottom-row entries are the quotient coefficients.

Step 5: Write the quotient with one lower degree than the original polynomial.

$Q(x) = 2x - 5, \quad R = -5$

Because the remainder is not 0, the divisor is not a factor of the polynomial.

Teacher answers & working

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E3

Divide $P(x) = 4x^3 - 4x^2 - 4x - 8$ by $x - 2$.

Answer:

$4x^3 - 4x^2 - 4x - 8 = (x - 2)\left(4x^2 + 4x + 4\right)$

Working:

Step 1: Convert the divisor to its root. The divisor is x - 2, so set it equal to zero and get $x = 2$.

Step 2: Write the coefficients of $P(x) = 4x^3 - 4x^2 - 4x - 8$ in descending powers of x. Include any zero coefficients if a power is missing.

Step 3: Carry out synthetic division: bring the first coefficient down, multiply by the root, then add down the next column.

2	4	−4	−4	−8
		8	8	8
	4	4	4	0

Step 4: The final entry in the bottom row is the remainder. The other bottom-row entries are the quotient coefficients.

Step 5: Write the quotient with one lower degree than the original polynomial.

$Q(x) = 4x^2 + 4x + 4, \quad R = 0$

Because the remainder is 0, the divisor is a factor of the polynomial.